POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Signal Analysis and Processing (New)

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary code words representing samples of an input signal The link is operated at 10,000 bits/sec and each input sample is quantized into 512 different voltage levels. 8

   i. What is the sampling frequency and folding frequency?
   ii. What is the Nyquist rate for the signal?
   iii. What are the frequencies in resulting discrete time signal x(n)?
   iv. What is the resolution?

   b) Determine the convolution sum of two sequences using graphical method: 7

$$x(n) = \{4,2,1,3\}, h(n) = \{1,2,2,1\}$$

2. a) Determine the z-transform and ROC of $x(n) = 5^n u(n)$ and $x(n) = cos(wn)u(n)$. 8

OR

Define Z transform and explain the properties of ROC of Z transform.

   b) An LTI system is described by the equation 7

$$y(n) = x(n) + 0.81x(n-1) - 0.81x(n-2) - 0.45y(n-2)$$

Determine the transfer function of the system. Sketch the poles and zeros on the z-plane. Assess the stability.

3. a) Realize the following system by its direct form I and form II structure. 7

$$y[n] = x[n] - 2x[n-1] + 0.5x[n-2] + 0.8y[n-2]$$

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b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure. 8

$$H(Z)=1+\frac{13}{24}z^{-1}+\frac{5}{8}z^{-2}+\frac{1}{3}z^{-3}$$

4. a) Determine H(z) using Impulse invariant technique for the analog system function. 7

$$H_a(s)=\frac{1}{(s+0.5)(s^2+0.5s+2)}$$

b) Design a low pass Butterworth digital Filter to give response of 3dB or less for frequencies upto 2kHz and attenuation of 20dB or more beyond 3kHz. Use the Bilinear transformation technique and obtain H(z) of the desired filter. Take sampling frequency as 8kHz. 8

OR

Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.9 \le |H(\omega)| \le 1;\ 0 \le \omega \le \frac{\pi}{2}$$

$$|H(\omega)| \le 0.2;\ \frac{3\pi}{4} \le \omega \le \pi$$

with $T = 1$ s.

5. a) How FIR digital filters are designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7

b) Design a linear FIR filter using Kaiser window to meet the following specifications: 8

$$0.99 \le |H(e^{jw})| \le 1.01;\ \text{for } 0 \le |w| \le 0.19\pi$$

$$|H(e^{jw})| \le 0.01;\ \text{for } 0.21\pi \le |w| \le \pi$$

6. a) Find the 8-point DITFFT of x(n) = {2, 2, 2, 2, 1, 1, 1, 1}. 8

b) Find the circular convolution of $x_1(n) = \{1,2,1,2\}$ and $x_2(n) = \{4,3,2,1\}$ by the graphical method. 7

7. Write short notes on: (**Any two**) 2×5

a) Properties of system

b) Gibbs phenomena and its remedy

c) Chebyshev Vs elliptic filter

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